Docker Screenshots:

1. docker compose up

```
(venv) blessyann@Mac fastapi_calc_sql % docker compose up
 [+] up 4/4
  ✔ Network fastapi_calc_sql_app-network Created                                                    0.0s
  ✔ Container postgres_db                Created                                                    0.1s
  ✔ Container fastapi_calculator         Created                                                    0.0s
  ✔ Container pgadmin                    Created                                                    0.1s
 Attaching to fastapi_calculator, pgadmin, postgres_db
 Container postgres_db Waiting
 postgres_db  |
 postgres_db  | PostgreSQL Database directory appears to contain a database; Skipping initialization
 postgres_db  |
 postgres_db  | 2026-04-07 23:32:02.517 UTC [1] LOG:  starting PostgreSQL 15.17 (Debian 15.17-1.pgdg13+1) on aarch64-unknown-linux-gnu, compiled by gcc
 (Debian 14.2.0-19) 14.2.0, 64-bit
 postgres_db  | 2026-04-07 23:32:02.518 UTC [1] LOG:  listening on IPv4 address "0.0.0.0", port 5432
 postgres_db  | 2026-04-07 23:32:02.518 UTC [1] LOG:  listening on IPv6 address "::", port 5432
 postgres_db  | 2026-04-07 23:32:02.519 UTC [1] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
 postgres_db  | 2026-04-07 23:32:02.524 UTC [29] LOG:  database system was shut down at 2026-04-04 23:16:02 UTC
 postgres_db  | 2026-04-07 23:32:02.534 UTC [1] LOG:  database system is ready to accept connections
 fastapi_calculator  | INFO:     Will watch for changes in these directories: ['/app']
 fastapi_calculator  | INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
 fastapi_calculator  | INFO:     Started reloader process [1] using StatReload
 fastapi_calculator  | INFO:     Started server process [8]
 fastapi_calculator  | INFO:     Waiting for application startup.
 fastapi_calculator  | INFO:     Application startup complete.
 Container postgres_db Healthy
 pgadmin             | postfix/postlog: starting the Postfix mail system
 pgadmin             | [2026-04-07 23:32:17 +0000] [1] [INFO] Starting gunicorn 23.0.0
 pgadmin             | [2026-04-07 23:32:17 +0000] [1] [INFO] Listening at: http://[::]:80 (1)
 pgadmin             | [2026-04-07 23:32:17 +0000] [1] [INFO] Using worker: gthread
 pgadmin             | [2026-04-07 23:32:17 +0000] [88] [INFO] Booting worker with pid: 88
```

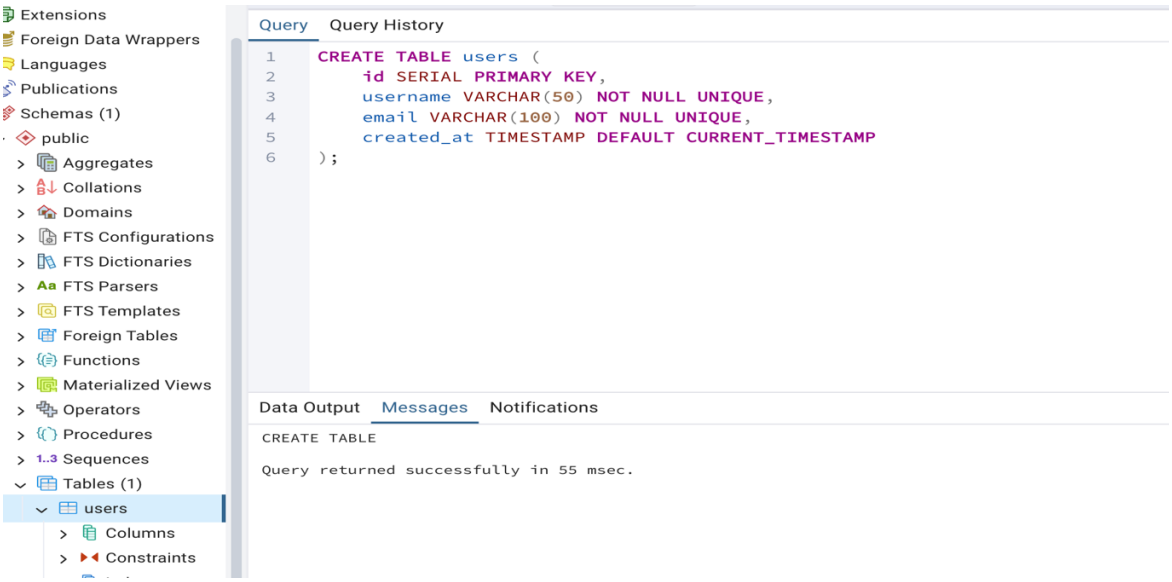
2. docker ps

```
blessyann@Mac fastapi_calc_sql % docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED              STATUS                         PORTS
    NAMES
8085b9c3e6cb   dpage/pgadmin4         "/entrypoint.sh"         About a minute ago   Up About a minute              0.0.0.0:5050->80/tcp, [::]:5050->80/tcp
    pgadmin
0d00e59794a3   fastapi_calc_sql-web   "uvicorn main:app --…"   About a minute ago   Up About a minute (unhealthy)  0.0.0.0:8000->8000/tcp, [::]:8000->8000/tc
p   fastapi_calculator
6813f8cf18fc   postgres:15            "docker-entrypoint.s…"   About a minute ago   Up About a minute (healthy)    0.0.0.0:5432->5432/tcp, [::]:5432->5432/tc
p   postgres_db
```

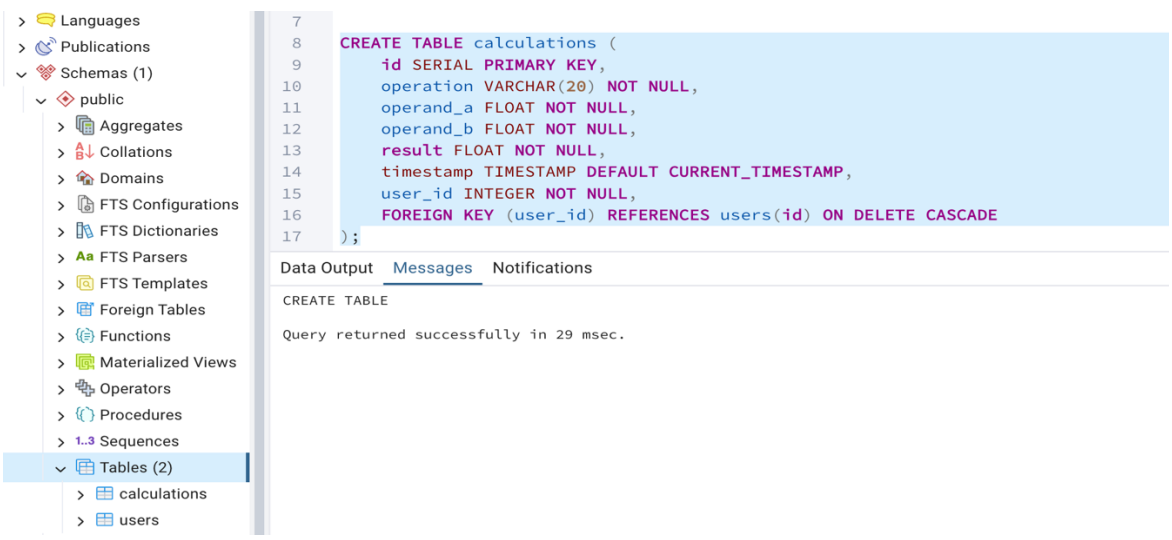
pgAdmin SQL queries:

Table creation:

1. Table ‘users’ created and ‘id’ is set as primary key.

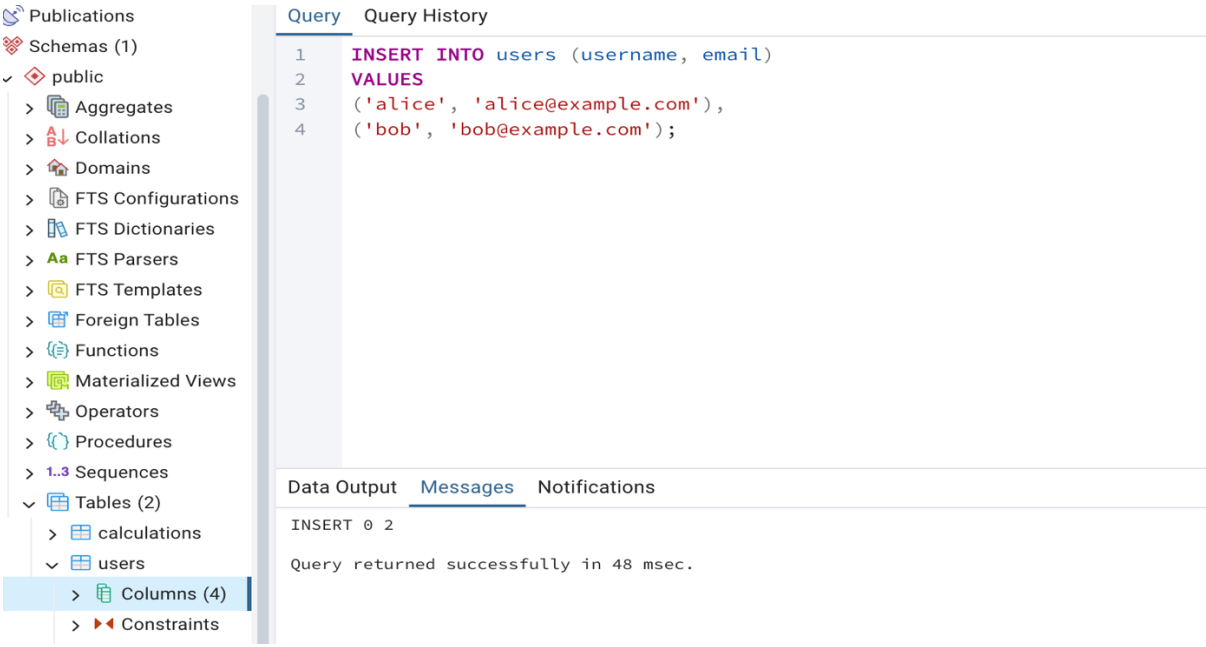


2. Table ‘calculations’ created; ‘id’ is set as primary key and ‘user_id’ as foreign key referencing users(id).

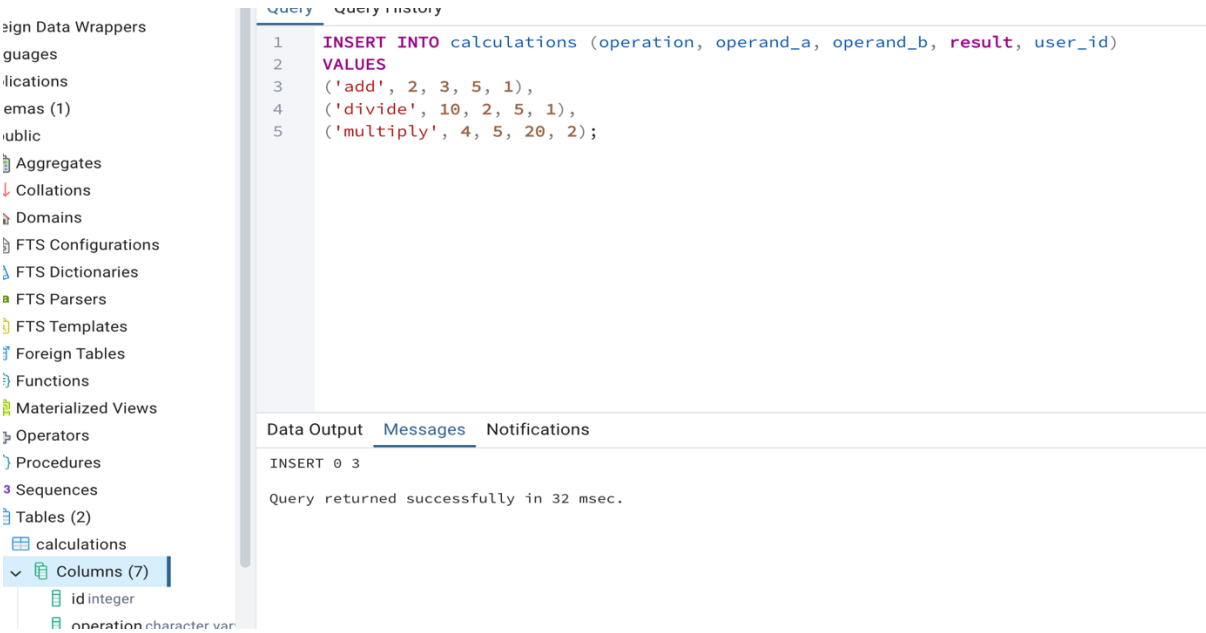


Insert Records:

3. Into the ‘users’ table 2 rows are inserted with username and email.



4. Into ‘calculations’ table 3 rows with operation, 2 operands (a and b), result and corresponding user_id are inserted.



Query Data:

5. Retrieve all ‘users’ table rows using the selected query

Query | Query History

```sql
-- Retrieve all users
SELECT * FROM users;

-- Retrieve all calculations
SELECT * FROM calculations;

-- Join users and calculations
SELECT u.username, c.operation, c.operand_a, c.operand_b, c.result
FROM calculations c
JOIN users u ON c.user_id = u.id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 2

	id [PK] integer	username character varying (50)	email character varying (100)	created_at timestamp without time zone
1	1	alice	alice@example.com	2026-04-04 23:02:42.543052
2	2	bob	bob@example.com	2026-04-04 23:02:42.543052

6. Retrieve all ‘calculations’ table rows using the selected query

Query Query History Scratch Pad

```sql
-- Retrieve all users
SELECT * FROM users;

-- Retrieve all calculations
SELECT * FROM calculations;

-- Join users and calculations
SELECT u.username, c.operation, c.operand_a, c.operand_b, c.result
FROM calculations c
JOIN users u ON c.user_id = u.id;
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1

	id [PK] integer	operation character varying (20)	operand_a double precision	operand_b double precision	result double precision	timestamp timestamp without time zone	user_id integer
1	1	add	2	3	5	2026-04-04 23:05:26.582007	1
2	2	divide	10	2	5	2026-04-04 23:05:26.582007	1
3	3	multiply	4	5	20	2026-04-04 23:05:26.582007	2

7. JOIN combines data from both tables, showing which user performed each calculation

Query Query History

```sql
-- Retrieve all users
SELECT * FROM users;

-- Retrieve all calculations
SELECT * FROM calculations;

-- Join users and calculations
SELECT u.username, c.operation, c.operand_a, c.operand_b, c.result
FROM calculations c
JOIN users u ON c.user_id = u.id;
```

Data Output Messages Notifications

Showing rows: 1 to 3

	username character varying (50)	operation character varying (20)	operand_a double precision	operand_b double precision	result double precision
1	alice	add	2	3	5
2	alice	divide	10	2	5
3	bob	multiply	4	5	20

Update Record:

8. Updated a calculation result, demonstrating modification of existing data.

Query Query History

```
UPDATE calculations
SET result = 6
WHERE id = 1;
```

Data Output Messages Notifications

```
UPDATE 1

Query returned successfully in 35 msec.
```

Delete Record:

9. Deleted a record from the calculations table.

Query Query History

```
DELETE FROM calculations
WHERE id = 2;
```

Data Output Messages Notifications

```
DELETE 1

Query returned successfully in 42 msec.
```